

Seizures

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Synonym(s): Convulsion, Fit

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Introduction

- Definition:** the term **seizure** literally means a sudden attack or recurrence of disease and as such the term is non-specific. It is also used to describe an epileptic seizure and is often used interchangeably with the word ‘convulsion’.
- Epileptic seizure:** the physical manifestation of paroxysmal transient disturbance of central nervous system function resulting from excessive and/or hypersynchronous abnormal neuronal activity within the cerebral cortex.
- Cause:** a seizure should be considered as a sign of forebrain disease rather than a diagnosis. Causes for seizures are therefore numerous but can be broadly grouped into extracranial and intracranial causes.
- Signs:** seizures are often associated with autonomic signs such as urination, salivation and defecation.
- Diagnosis:** the key to diagnosis of a seizure is by inspection of an episode. Nowadays this can easily be achieved by the owner filming a typical episode.
- Treatment:** this centers on treatment of the primary cause when possible as well as the administration of antiepileptic medications as appropriate.
- Prognosis:** epileptic seizures have a varying prognosis depending on the primary cause.

 Follow the diagnostic tree for Intracranial vs Extracranial Seizures [Intracranial vs extracranial seizures](#).

Clinical tip:
Question: What is the most common form of generalized seizure in dogs?
Answer: Generalized tonic-clonic seizures.

Presenting signs

- Involuntary muscle contractions.
- Autonomic signs:
 - Salivation.
 - Urination.
 - Defecation.
- Behavioral signs.

Classification

- Epileptic seizures can be classified into two major categories:
 - Generalized Seizures:** these tend to be the most common seizure type in dogs. Generalized seizures have no localizing signs and indicate involvement of both cerebral hemispheres. Consciousness/awareness is impaired and motor manifestations are bilateral.

consciousness whilst suddenly falling to the ground and show chumping, chewing, foaming at the mouth, paddling of the legs, and sometimes the passing of urine or stools. They usually last no more than a few minutes.

- **Myoclonic seizures** are generalized seizures by definition in that they involve both cerebral hemispheres and involve loss of consciousness. However, they are often so brief in nature that an objective measurement of consciousness is impossible and observation of an episode may pass without any obvious discernible loss of awareness. Myoclonus manifests as a sudden jerk as if the dog has been given an electric shock. Therefore, the keywords in identifying myoclonus are 'shock-like' movements. When myoclonus occurs in series, the resulting jerks may be synchronous or moderately asynchronous.
- **Absence seizures** are another type of generalized seizure. This is when a dog loses awareness of their surroundings for a transient period of time. Dogs will seem to stare vacantly into space and not respond to their name being called. These seizures are very uncommon and are sometimes referred to as petit mal seizures.
- **Partial seizures** [Partial seizures](#): these tend to be more common in cats than in dogs. This type of seizure indicates abnormal neuronal activity in a localized region of the cerebral hemisphere. Any portion of the body can be involved during a focal seizure depending on the region of the brain affected. There are two main forms of partial seizure:
 - **Simple partial seizure**: unaltered consciousness with asymmetric localized motor signs such as facial twitching, or the clonus of muscle groups of one limb ('deer-stalking').
 - **Complex partial seizure**: these differ from simple partial seizures in that they involve some degree of impaired consciousness/awareness. They include psychomotor seizures which are 'behavioral' seizures involving the limbic system which may present as rage, aggression without provocation, fly-catching, running in circles, floor licking, vocalisation, tail chasing, star-gazing etc. Psychomotor seizures are controversial in the sense that they may represent a form of obsessive compulsive disorder. No evidence exists to support either view strongly.
- A seizure may start in a focal region of the brain only to spread throughout both cerebral hemispheres, resulting in a focal seizure with secondary generalization.

Acute presentation

- Status epilepticus [Status epilepticus](#).

Age predisposition

- **<6 months:**
 - Hepatic encephalopathy [Hepatic encephalopathy](#) (eg portosystemic shunt [Congenital portosystemic shunt \(CPSS\)](#)).
 - Juvenile onset hypoglycemia.
 - Congenital malformations (eg hydrocephalus [Hydrocephalus](#) , porencephaly/hydranencephaly).
 - Non-infectious meningoencephalitis [Encephalitis](#) (eg meningoencephalitis of unknown origin).
 - Infectious meningoencephalitis (eg neosporosis, canine distemper virus and Angiostrongylus infestation).
- **6 months-6 years:**
 - Idiopathic epilepsy most likely [Epilepsy: idiopathic](#).
- **>6 years:**
 - Hypoglycemia (eg insulinoma).
 - Neoplasia [Brain: neoplasia](#) (eg glioma, meningioma, etc).
 - Cerebrovascular accident [Intracranial hemorrhage](#) (eg middle or rostral cerebral arterial infarct).

Breed/Species predisposition

- Idiopathic epilepsy [Epilepsy: idiopathic](#) is common in some breeds:
 - Labrador [Retriever: Labrador](#).
 - Golden Retriever [Retriever: Golden](#).
 - English Springer Spaniel [English Springer Spaniel](#).
 - Border Collie [Border Collie](#).

- German shepherd dog [📖 German Shepherd Dog](#).
- Breed predispositions exist for certain pathologies:
 - Brachycephalic breeds → brain tumor [📖 Brain: neoplasia](#) (ie glioma [📖 Astrocytoma](#)).
 - Toy breeds, eg Yorkshire Terrier [📖 Yorkshire Terrier](#) → hydrocephalus [📖 Hydrocephalus](#).
 - Staffordshire Bull Terrier [📖 Staffordshire Bull Terrier](#) → hydroxyglutaric aciduria [📖 L-2 - hydroxyglutaric aciduria](#).

Cost considerations

- Cross-sectional imaging of the brain, if required, may be expensive.
- Long-term antiepileptic medication may be required.

Special risks

- Refractory epilepsy will often result in a decision for euthanasia.
- Death can occur during status epilepticus [📖 Status epilepticus](#).
- Discontinuing antiepileptic medication abruptly may trigger ‘abstinence syndrome’. This involves central nervous system hyperexcitability; motor, autonomic and behavioral changes; and potentially withdrawal seizures.

Pathogenesis

Etiology

Intracranial (ie structural/symptomatic or functional)

- Idiopathic epilepsy [📖 Epilepsy: idiopathic](#).
- Congenital diseases, eg Hydrocephalus [📖 Hydrocephalus](#), porencephaly/hydranencephaly, storage diseases [📖 Storage disease](#), arachnoid cyst [📖 Spine: arachnoid cyst](#).
- Infectious meningoencephalitis.
- Non-infectious meningoencephalitis (ie meningoencephalitis of unknown origin such as granulomatous meningoencephalitis [📖 Granulomatous meningoencephalomyelitis](#)).
- Trauma [📖 Brain: trauma](#).
- Central nervous system neoplasia [📖 Brain: neoplasia](#) (eg glioma, meningioma, oligodendroglioma).
- Nutritional (eg thiamine deficiency).
- Vascular (eg cerebrovascular accident such as infarct or hemorrhage).

Extracranial (ie metabolic, toxic or anoxic seizures)

- Metabolic disease:
 - Hypoglycemia [📖 Hypoglycemia](#) (eg insulinoma, sepsis, hepatic insufficiency, hypoadrenocorticism, insulin overdose, juvenile hypoglycemia, xylitol toxicity, paraneoplastic cause).
 - Hypocalcemia [📖 Primary maternal dystocia](#) (eg eclampsia, renal failure, pancreatitis, ethylene glycol intoxication, primary hypoparathyroidism).
 - Hypernatremia [📖 Hypernatremia](#) (ie excess intake or renal/gastrointestinal loss).
 - Hepatic encephalopathy [📖 Hepatic encephalopathy](#) (eg portosystemic shunt or severe hepatic insufficiency).
 - Uremia [📖 Uremia](#).
 - Hyperlipoproteinemia [📖 Blood biochemistry: total lipid](#) (eg familial or secondary to hypothyroidism).
 - Hypertensive encephalopathy.
 - Polycythemia.
- Toxicity:
 - Lead [📖 Lead toxicity](#).
 - Metaldehyde [📖 Metaldehyde poisoning](#).
 - Ethylene glycol [📖 Ethylene glycol poisoning](#).
 - Strychnine [📖 Strychnine toxicity](#).
 - Chocolate [📖 Theobromine / chocolate poisoning](#).
 - Alpha-Chloralose.

- Chlorinated hydrocarbons  [Carbamate poisoning](#).
- Xylitol  [Xylitol toxicity](#).
- Anoxic:
 - Hypoxia, eg cardiac  [Heart: congestive heart failure](#) or respiratory insufficiency.

Pathophysiology

- Epilepsy can be caused by an intracranial (ie congenital or acquired brain damage) or extracranial problem (ie a problem with the content or supply of blood to the brain) .
- The normal brain cell maintains an unevenly distributed electrical charge across the cell membrane. The interior of the cell is negative with respect to the exterior, and this difference is maintained in the resting state primarily via the Na⁺-K⁺ ATPase pump that removes three sodium ions in exchange for two potassium ions into the cell.
- The **resting potential** of the neuron refers to the difference between the voltage inside and outside the neuron.
- The resting potential of the average neuron is around -70 millivolts, indicating that the inside of the cell is 70 millivolts less than the outside of the cell.
- When the cell is excited, the sodium channels open and positive sodium ions surge into the cell. Once the cell reaches a certain threshold (depolarisation), an action potential will fire, sending an electrical signal down the axon.
- After the neuron has fired, there is a refractory period in which another action potential is not possible.
- During this time, the potassium channels open and the sodium channels close, gradually returning the neuron to its resting potential. Once the neuron has returned to the resting potential, it is possible for another action potential to occur.
- There are neurons that are either excitatory or inhibitory.
- The **excitation of neurons** is mainly mediated by the **glutamate neurotransmitters** (also aspartate and acetylcholine) and their receptors – creating excitatory post-synaptic potentials (EPSPs)
- The **inhibition of neurons** is mediated by the **GABA neurotransmitter** (γ-aminobutyric acid; also glycine, taurine and noradrenaline) and their receptors - creating inhibitory post-synaptic potentials (IPSPs).
- The neuronal membrane potential is determined by the balance of EPSPs and IPSPs - if this balance is compromised, an epileptic seizure will result.
- The basic pathophysiological processes that result in seizures are excessive excitation or loss of inhibition (disinhibition):
 - Hypoglycemia → loss of energy substrate for the Na⁺-K⁺ ATPase pump, failure to extrude Na⁺, increasing cell positivity resulting in depolarisation (excessive excitation).
 - In a disease process where inhibitory transmitters are unable to function (eg hepatic encephalopathy), the lack of inhibition allows for unregulated depolarisation.
- Two interesting phenomena that occur due to seizure activity include:
 - Mirror focus - where a seizure focus creates similar activity in a homologous area of the contralateral hemisphere.
 - Kindling - where one seizure increases the likelihood of further seizures. With time both mirror foci and kindled foci may become autonomous and form a new, independent seizure focus.
- Why seizures terminate as rapidly as they begin is not known. Metabolic exhaustion of neurons is not an adequate explanation. Extracortical inhibitory centers, such as within the cerebellum, may play a role. Ablations of the cerebellum, for example, facilitate seizure activity. Phenytoin, a commonly used antiepileptic medication in humans, dramatically increases the rate of firing of Purkinje neurons. Other areas such as the caudate and parts of the thalamus and reticular formation may also help to terminate seizure activity.
- It is often noted that seizures occur in the middle of the night in dogs. One explanation suggests that during low levels of awareness, drowsiness and dreamless sleep, decreased activity in the reticular formation allows for reverberating circuits between the thalamus and the cortex to synchronise. Additionally, groups of neurons which are only mildly hyperactive in the awake state become excitable and fire consistently during sleep.

Diagnosis

Presenting problems

- Seizures.

Client history

- It is important to confirm that the dog is definitely having a seizure, and if it is, to ask questions to ascertain whether an intracranial (structural/symptomatic or functional) or extracranial cause (metabolic, toxic or anoxic) is likely:
 - Age at the onset of the first seizure (enables a narrowing of the list of likely differential diagnoses as different diseases are more likely at different ages).
 - Asking the owner to obtain video footage of an episode is invaluable as invariably these dogs will present with no abnormalities on the day of assessment meaning recognition of the disorder can only be based on the owner's description.
 - Ask for a description of the episodes verbatim:
 - Any autonomic signs?
 - If autonomic signs (eg urination, defecation and salivation) are observed in conjunction with paroxysmal involuntary movements then an epileptic seizure is highly likely.
 - Can the dog respond to the owner during an episode?
 - Loss of awareness is very common with epileptic seizures although partial seizures can occur with no loss of consciousness. If a dog has involuntary movements of all four legs during an episode with normal consciousness then this would suggest a non-epileptic seizure event as a generalized tonic-clonic seizure can only occur with loss of consciousness.
 - Duration of the episode?
 - Epileptic seizures are usually less than 5 minutes in duration. If a long duration is described (eg 30 minutes to one hour or more) then this would suggest an epileptic seizure is less likely, particularly when accompanied with no postictal signs.
 - Are there any post-ictal signs?
 - The lack of post-ictal signs following a long paroxysmal episode suggests an epileptic seizure is less likely. However, short epileptic seizures (< 5 minutes) can occur without obvious post-ictal signs.
 - Do all episodes look the same?
 - Epileptic seizures are usually stereotypical (ie each event looks the same).
 - Does the dog go floppy during an episode?
 - The presence of decreased tone during a paroxysmal episode is strongly suggestive of a non-epileptic seizure disorder.
- Frequency of seizures (the aim of antiepileptic treatment is not to cure the animal of epilepsy but to "control" the seizures with "acceptable" side-effects). The decision to start treatment should be based on the frequency of the seizures:
 - There is no correlation between seizure frequency and the likely underlying disease process as an animal with idiopathic epilepsy might experience seizures on a weekly basis or present in status epilepticus while an animal in the early stage of a brain tumor might be presented with only one recorded seizure event.
- What was the animal doing just before the episode occurred (dogs with idiopathic epilepsy typically seizure when they are at rest or sleeping - seizures at exercise or associated with excitement are more common with cardiovascular disease or metabolic disease, eg hypoglycemia).
- Aside from the seizures is the dog normal?
 - Any abnormal inter-ictal behavior is indicative of a structural/symptomatic forebrain lesion or metabolic cause for the seizures. Clinical signs of forebrain disease include:
 - Altered mentation.
 - Blindness.
 - Relentless pacing/circling.
 - Loss of learned behavior (eg toilet training).
 - Following an epileptic seizure, transient forebrain signs lasting up to 24 hours may be present. Therefore if these signs are observed, ensure they disappear 24 hours after the last seizure was observed.

- In some cases, the lesion causing the seizures lies in an otherwise 'silent' region of the brain (causing only seizures but no other localizing neurological deficits) - this most commonly occurs in the olfactory or prefrontal lobes. During the early stages of a slowly enlarging mass within one of these silent regions only seizures may be evident but with time other neurological deficits reflecting the site of the will develop).
- Previous medical history (sudden cessation of antiepileptic medication can trigger seizure activity, hepatotoxic drugs can result in liver damage and hepatic encephalopathy, some drugs have neurological side-effects).
- Vaccination status (some infectious viral diseases can result in seizures but are prevented by vaccination).
- Travel history (some diseases potentially causing seizures are more common or only present abroad).
- Any familial history of seizures (epilepsy may be proven or suspected to be inherited in some breeds, eg Labrador Retriever, Golder Retriever, Border Collie, German Shepherd dog).

Clinical signs

- An epileptic seizure is not a disease entity in itself but a clinical sign generally indicative of a forebrain disorder.
- Seizure etiology can be classified as intracranial or extracranial.
- Intracranial causes are further subdivided into those where:
 - A structural (symptomatic) lesion is identified (vascular, inflammatory/infectious, traumatic, congenital, neoplastic disease) and
 - Those where no such lesion is present, that is primary (functional or idiopathic epilepsy).
- Features of extra-cranial seizures:
 - Metabolic seizures:
 - Multifocal neurological examination - ie neurological deficits relate to the forebrain plus other components of the nervous system, eg the neuromuscular system.
 - Symmetrical neurological signs most commonly seen.
 - Inter-ictal signs are usually present that can wax and wane with dogs having 'good' and 'bad' days.
 - Toxic seizures:
 - Seizures are not usually recurrent, occurring during a discrete time period.
 - Myoclonus and twitching are common features.
 - Often accompanied with gastrointestinal signs, eg vomiting and diarrhea.
 - Anoxic seizures:
 - Associated cardiorespiratory signs, eg arrhythmias.
 - Associated triggers leading to increased vagal tone, eg excitement/stimulation (eg feeding), coughing, regurgitation, defecation.
- Features of intracranial seizures:
 - Symptomatic seizures:
 - Altered mentation.
 - Blindness.
 - Relentless pacing/circling.
 - Loss of learned behavior (eg toilet training).
 - Idiopathic seizures:
 - Onset between 6 months and 6 years of age.
 - Normal inter-ictally.
 - Recurrent seizures.
 - Breed predisposition.

Diagnostic investigation

Urinalysis

- Specific gravity [📖 Urinalysis: specific gravity.](#)
- Dipstick assessment [📖 Urinalysis: dipstick.](#)

Hematology

- Packed cell volume [\[📖 Hematology: packed cell volume\]](#) , for polycythemia [\[📖 Polycythemia: secondary\]](#).

Biochemistry

- Routine screen [\[📖 Blood biochemistry: overview\]](#) :
 - Assess liver function via fasting sample (AST [\[📖 Blood biochemistry: aspartate aminotransferase \(AST\)\]](#) , ALP [\[📖 Blood biochemistry: alkaline phosphatase \(ALP\)\]](#) , ALT [\[📖 Blood biochemistry: alanine aminotransferase \(SGPT, ALT\)\]](#) , and bile acid stimulation test [\[📖 Blood biochemistry: bile acids\]](#) (to rule out hepatic encephalopathy):
 - Urea [\[📖 Blood biochemistry: urea\]](#).
 - Cholesterol [\[📖 Blood biochemistry: cholesterol\]](#).
 - Albumin [\[📖 Blood biochemistry: albumin\]](#).
 - Glucose [\[📖 Blood biochemistry: glucose\]](#).
 - Bile acid stimulation test [\[📖 Blood biochemistry: bile acids\]](#).
 - Ammonia [\[📖 Blood biochemistry: ammonia\]](#) (if available).
 - Assess renal function (to rule out uremic encephalopathy):
 - Urea [\[📖 Urinalysis: urea\]](#).
 - Creatinine [\[📖 Urinalysis: creatinine\]](#).
 - Fasting glucose (to rule out hypoglycemia).
 - Ionized calcium [\[📖 Blood biochemistry: ionized calcium\]](#) (to rule out hypocalcemia).
- Blood lead levels if appropriate.

Cytopathology

- Cytological examination of cerebrospinal fluid (CSF) [\[📖 Cerebrospinal fluid: cytology\]](#) may show evidence of inflammatory response.
- CSF results should be interpreted in light of neurological examination [\[📖 Neurological examination\]](#) , clinical findings and other ancillary aids, eg MRI.

 CSF results alone are poorly sensitive and not specific.

Radiography

- Thoracic [\[📖 Radiography: thorax\]](#) and abdominal radiographs [\[📖 Radiography: abdomen\]](#) to rule out underlying cause.
-  Skull radiographs are rarely helpful.

Serology

- Toxoplasma [\[📖 Toxoplasma antibody titer\]](#) and neospora titers if suspicious based on cross-sectional imaging findings.
- When performed alone these serological tests offer little clinical value.

Other

- Blood pressure measurement (to rule out hypertensive encephalopathy).
- **MRI** [\[📖 Magnetic resonance imaging: brain\]](#) or **CT** [\[📖 Computed tomography\]](#) for structural brain damage (vascular, inflammatory infectious, neoplastic, anomalous, traumatic).
- **ECG** [\[📖 ECG: overview\]](#) for cardiac abnormalities.
- EEG (electroencephalography) is rarely useful.

Confirmation of diagnosis

Discriminatory diagnostic features

- Video footage of an episode.
- Categorising the type of seizure, eg generalized (tonic-clonic or myoclonic) or partial (complex or partial).
- After establishing the general categories of seizure causes, specific causes pertinent to the

individual case should be considered.

Definitive diagnostic features

- Identification of an underlying cause.

Differential diagnosis

- Important to exclude other diseases that mimic seizures:
- Syncope (cardiac disease) [📖 Syncope](#).
- Weakness (hypoglycemia, myasthenia gravis [📖 Myasthenia gravis](#)).
- Movement disorders (paroxysmal dyskinesia [📖 Paroxysmal dyskinesia](#), eg canine epileptoid cramping syndrome [📖 Canine epileptoid cramping syndrome](#) epidodic falling of the Cavalier King Charles spaniel [📖 Episodic falling \(Cavalier King Charles Spaniel\)](#), etc).
- Narcolepsy [📖 Narcolepsy](#) / Cataplexy.
- Myokymia and neuromyotonia.
- Myotonia [📖 Myotonic myopathy \(myotonia\)](#).
- Postural myoclonus (eg idiopathic head bobbing).
- Vestibular episode [📖 Vestibular disease: geriatric](#).

Treatment

Standard treatment

- Treat the underlying cause if possible [📖 Seizure: management](#).
- Initiate antiepileptic medication where appropriate.
- When to start treatment? This depends on several factors:
 - Frequency of seizures.
 - How well the owner copes with seizures.
 - Side effects of treatment.
 - Commitment to long term treatment accompanying monitoring protocols.
- When to change treatment or add additional therapy?
 - You should consider altering your treatment plan if:
 - Seizure frequency has not reduced by >50%.
 - Seizure frequency is >1 seizure per 6 weeks.
 - Cluster seizures or status epilepticus occur.

Monitoring

- Antiepileptic therapy requires monitoring:
 - 3-6 monthly biochemical and hematological testing is appropriate with all antiepileptic medication to ensure systemic adverse effects are not developing.
- Phenobarbital [📖 Phenobarbital](#) requires serum concentration testing [📖 Phenobarbital assay](#). This is indicated:
 - Steady state blood levels are achieved after starting treatment (10-15 days in dogs). This provides a baseline to further guide changes in doses according to clinical circumstances.
 - When the seizure frequency increases or the patient becomes refractory to the phenobarbital therapy.
 - Every 3-6 months to verify that the blood concentration has not drifted out of the intended range.
 - When drug-related side effects are suspected.

Subsequent management

Monitoring

- It may be possible to wean epileptic dogs off therapy very gradually if no seizures have been detected for a number of years. However, this will always run the risk of withdrawal seizures and so owners should be well informed of this risk before undertaking such action.

Outcomes

Prognosis

- Approximately 60-70% of idiopathic epileptics will have their seizure frequency or severity decreased with the currently available antiepileptic medications.
- Up to 30% of dogs with idiopathic epilepsy can develop drug-resistant epilepsy (pharmacoresistance), ie they are poorly controlled despite adequate doses and serum concentrations of antiepileptic medication.
- The prognosis for dogs with other structural or metabolic causes depends upon appropriate management of the underlying disease.

Expected response to treatment

- Reducing severity and/or frequency of seizures.
- Goals vary dependent on cause of epilepsy.
- Typically the following are cited as achievable goals when managing dogs with idiopathic epilepsy:
 - 50% reduction in seizure frequency.
 - Seizure frequency < 1 seizure per 6 weeks.
- Around 75% of dogs with idiopathic epilepsy will continue to experience seizures whilst on treatment.
- Around one third of dogs will remain poorly controlled despite suitable treatment.

Reasons for treatment failure

- Inadequate treatment (eg dosage is too low).
- Incorrect condition diagnosed (consider diseases that mimic epileptic seizures such as paroxysmal movement disorders).
- Unable to establish or treat underlying cause.
- Progressive cause, eg progressive cirrhosis, brain tumor.
- Owner compliance, eg not giving the prescribed dosage of medication.
- Pharmacoresistance (ie drug-resistant epilepsy) - occurs in one third of dogs with idiopathic epilepsy.
- A new disease exacerbating the seizures.

Further Reading

Publications

Refereed Papers

- Recent references from [PubMed](#) and [VetMedResource](#).
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